

California Home Wildfire Risk Analysis: Combatting LA Wildfires through an LLM-Powered Risk Assessment & Recommendation System for Homeowners & Homebuyers

Yvonne Lopez Davila
Tufts University
yvonne.lopez_davila@tufts.edu

Emily Yuan
Tufts University
Emily.yuan@tufts.edu

Abstract

California's escalating wildfire crisis has created an urgent need for accessible, property-level risk information, yet relevant data remains fragmented across disconnected sources. This paper presents the Wildfire Risk Analysis Tool (<https://wildfire-risk-report-generator.onrender.com/>), a full-stack, LLM-powered system deployed as a publicly accessible web application, that aggregates structured wildfire, insurance, and housing market data to generate tailored risk reports for California homeowners and homebuyers. The system is evaluated through an LLM-as-judge pipeline evaluation, a randomized controlled experiment, and a user feedback survey. Results indicate strong data grounding across structured services, with static information presentation identified as the primary area for improvement.

All relevant code is available at https://github.com/yvonne-lopez-davila/genAI_LA_wildfires, and a system demo is available at https://1drv.ms/v/c/ab3df13dbe14d7d7/IQB2kVfN9nJqSrIV-VWU686_AUBe7ijCWAUdCDZCnIcoWdI?e=tYTu2a.

1 Introduction

California's wildfire crisis has intensified dramatically in recent years, culminating in the January 2025 Los Angeles fires that destroyed thousands of properties and displaced tens of thousands of residents. The collapsing private insurance market has left homeowners increasingly dependent on the state-mandated FAIR Plan. The data that is required to navigate these risks exist but are scattered across sources with no unified platform for homeowners or buyers to query.

This paper presents the Wildfire Risk Analysis Tool, a full-stack, LLM-powered application that aggregates data from five structured services to generate tailored, role-specific wildfire risk reports for California properties. The system supports distinct paths for homeowners, who receive ranked mitigation recommendations grounded in structural damage data from nearby affected properties, and homebuyers, who receive prioritized financial and insurance risk insights relative to market trends over time. Reports are generated through a retrieval-augmented generation (RAG) pipeline and delivered through a web-based map interface. It is supplemented by a conversational report assistant. The system is evaluated

through an LLM-as-judge pipeline evaluation, a randomized controlled experiment, and a user feedback survey assessing output quality.

2 Background & Related Work

Several platforms have already incorporated property-level climate risk tools. Real estate sites such as Zillow, Redfin, and Realtor.com offer climate risk indicators, and the nonprofit CarbonPlan recently launched Open Climate Risk, a free app for assessing property-level climate exposure. Some state agencies, such as the Connecticut Insurance Department, provide numerical future risk ratings for properties. However, different platforms use varying methodologies, producing inconsistent risk scores for the same property, and FEMA hazard maps have been found to be frequently outdated. Researchers at the Environmental Defense Fund and the Wharton School study how wildfire risk necessitates trade-offs in the home-buying process and impacts long-term affordability through insurance premiums. First Street has estimated a \$1.47 trillion reduction in real estate value over the next 30 years due to climate-driven warming. On the direct response side, the ALERTCalifornia statewide monitoring system integrates AI capabilities developed by Microsoft AI for Good Lab to detect early fire indicators from continuous camera feeds.

Despite these efforts, no existing platform unifies fire hazard, insurance, structural damage, and market data at the property level into a single, role-differentiated report. This work addresses that gap by aggregating unstructured and structured data across these sources and generating tailored outputs for homeowners and homebuyers through an LLM-powered pipeline.

3 System Design

The Wildfire Risk Analysis System is a full-stack, LLM-powered risk-assessment system that RAG retrieval and quantitative trend analysis to generate structured insights from structural home characteristics, geospatial data, historical fire activity data, insurance data, and home value trends. The application supports distinct, role-specific workflows for prospective home buyers and existing homeowners, enabling tailored, actionable steps through ranked insights and mitigation recommendations. The full system is deployed on the cloud platform Render, through which it is widely accessible to users.

The system features can be discussed in three main components: the user-facing layer, the API and services layer, and the risk client layer.

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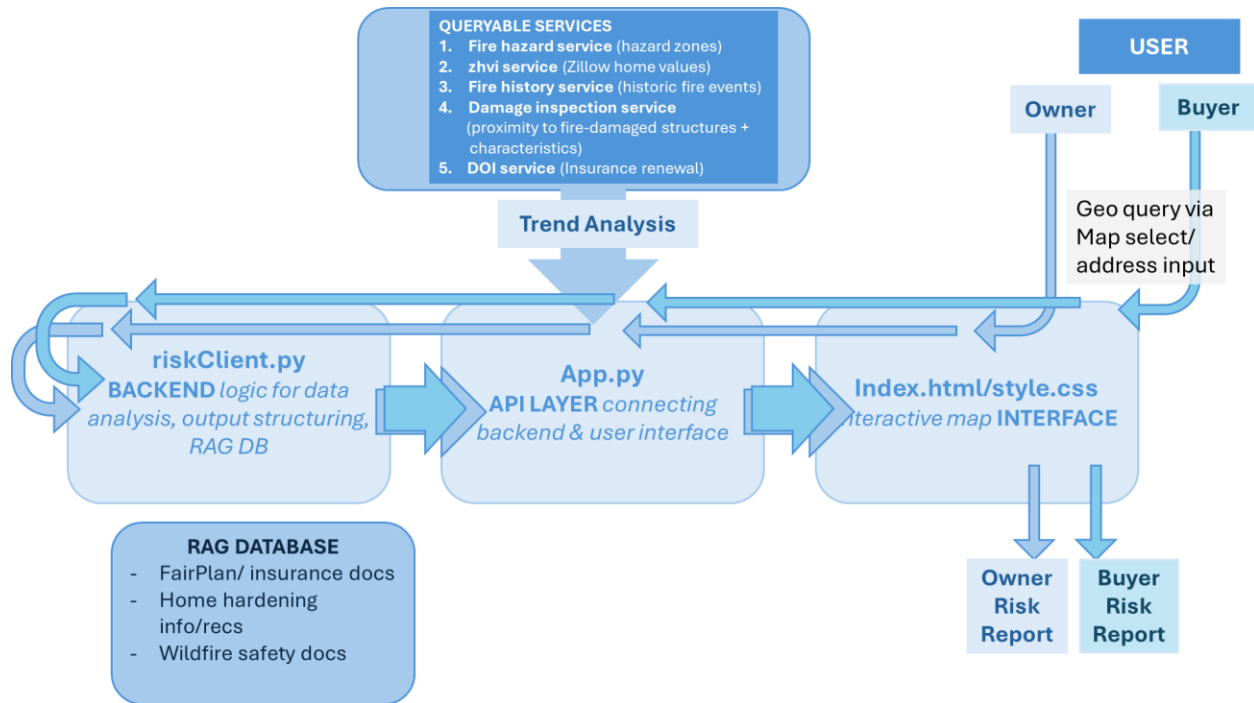


Figure 1: System architecture overview

3.1 Map Interface and Risk Report Generation

Users may interact with the tool through a web-based interface. First, the user selects his role as a current homeowner or a prospective buyer. Upon selection, this triggers the main interface, which features a queryable map; users can select directly from the map or find their desired address through the search bar. Users can then generate a tailored report for the selected address. While homeowner and homebuyer reports draw from the same information sources, there are several key differences in what and how information is displayed.

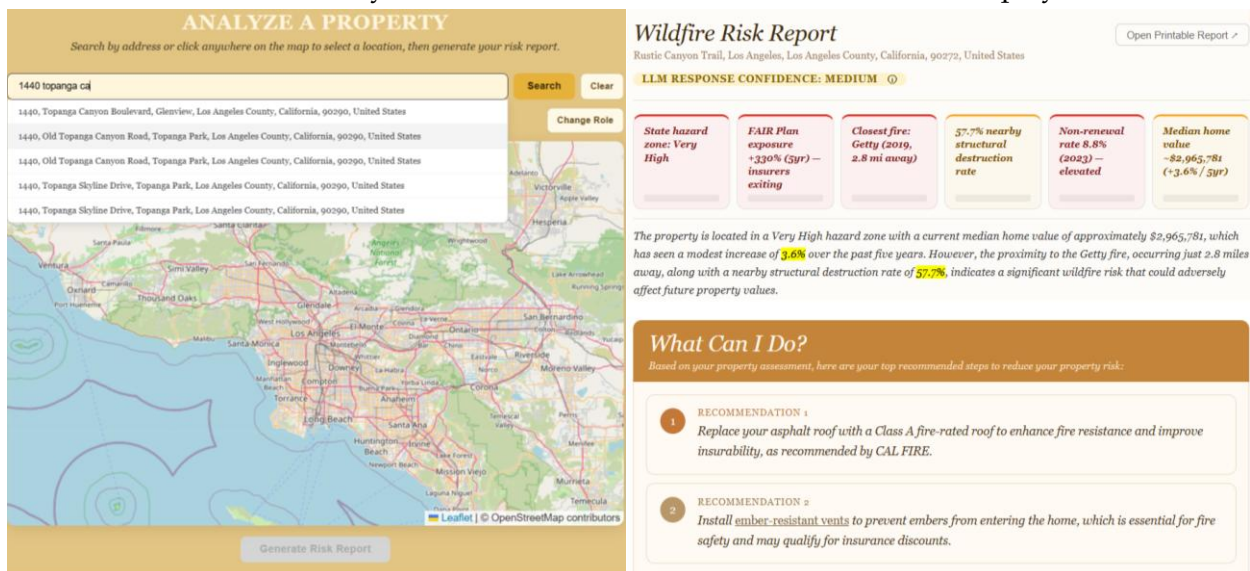


Figure 2: Map interface (left) and generated homebuyer report snapshot (right)

3.1.1 Homeowner Path

Homeowners are directed to an optional input page, which allows input of major structural characteristics: roof construction, the presence of enclosed or unenclosed eaves, and exterior siding. The homeowner path is designed to prioritize actionable insights: *What is my risk based on my home-specific characteristics, and what steps can I take to reduce this risk?* This is primarily achieved through ranked recommendations and direct comparison to the materials found in nearby structures that have been recently affected by damage from wildfires.

3.1.2 Homebuyer Path

Homebuyers are immediately directed to key risk insights, prioritized by urgency. The homebuyer path seeks to answer: *What are primary financial and safety risks relative to other zones in LA, and how have these risks changed over time?* Home value and market trends in relation to major wildfire events are prioritized, offering users guidance through significant relationships between pricing and fire data. This is followed by local insurability trends, which highlight withdrawal rates from private insurers and coverage details of the California state-mandated, last-resort insurance, Fair Access to Insurance Requirements (FAIR Plan). This report prioritizes highlighting trends and specific changes over time, allowing users to draw their own comparison insights.

3.1.3 Report Assistant Chatbot

The reports feature a wide variety of data insights and definitions that not all users may be familiar with. Thus, the system integrates a task-specific LLM that allows aids users in interpreting their report through conversational, natural language prompting. This is primarily intended to make the system more user-friendly and adaptable to users with varying backgrounds and experiences with statistical and risk-related terminologies.

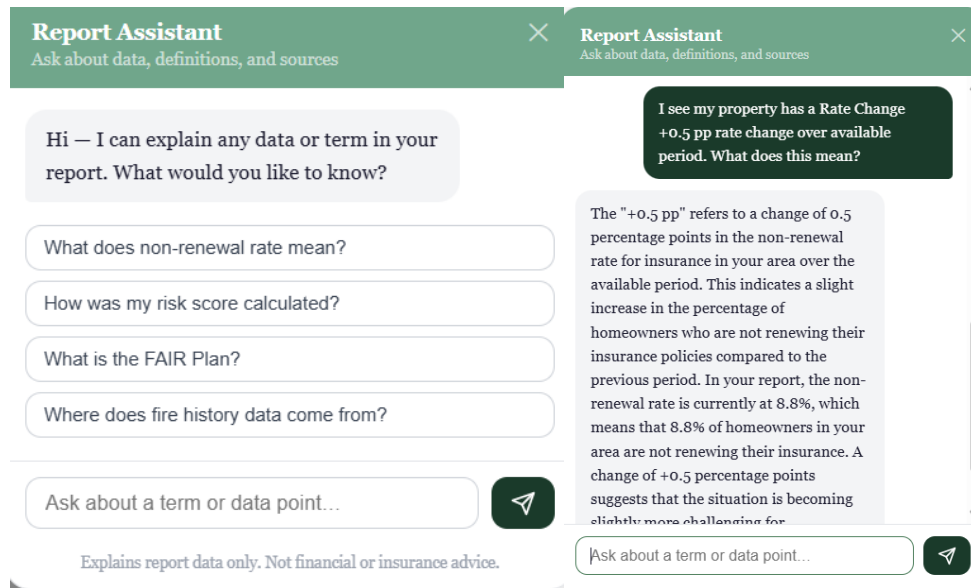


Figure 3: Report assistant interface supports user interaction with report data

3.1.4 Exportable Printing Feature

Generated reports may also be exported to independent tabs. This allows users to download and print reports as PDFs. This is primarily designed to enable comparison across different properties, as users may more easily compare risk reports across multiple property queries.

3.2 API Communication Layer

The interface connects with the core through an API layer built on a FastAPI framework, which supports two endpoints, `/analyzeFireRisk` and `/chat`.

3.2.1 Analyze Fire Risk Endpoint

The fire risk analysis endpoint is the primary channel through which user requests and relevant context are sent to the LLM core. Namely, the analysis endpoint sends back the queried location (latitude/longitude coordinates), extra context consisting of processed data from services, and the user role (owner vs buyer). Upon receiving LLM-powered insights, the API layer generates structures received backend information and sends it back to the interface layer, where it is styled into various graphics and modules.

3.2.1.1 Data Retrieval and Processing Services

The API layer draws from various information services, which are designed to perform objective, quantitative analyses and insights that are passed as context to the backend LLM-core. Services involve structured information retrieval and trend analysis aimed to quantify the level of impact across different variables. There are five distinct types of querying services that retrieve structured, zip code level information. The first involves home data trends; this service

draws trends from Zillow Home Value (ZHVI) and Zillow Observed Rent (ZORI) Indices, which measure market changes over time with respect to home value and property rent rates. Second, the fire hazard zone service retrieves the official, state-issued zone classification for the given zip code. Fire hazard severity zones (FHSZ) [1] are classified as moderate, high, and very high by the California State Fire Marshal. These zones reflect likelihood of fire based on vegetation, historical terrain data, and other such factors. The third retrieval service queries historical fire perimeter data [2], finding information and trends for major recent and nearby wildfire events. The fourth service relates to damage inspection data (DINS) [3]. This service retrieves and analyzes property characteristics and destruction rates of structures most strongly affected by wildfires within proximity of the property being assessed by the user. Lastly, the fifth service retrieves data from the California Department of Insurance. This includes FAIR Plan coverage and non-renewal rates and dropout rates of private insurers.

The outputs of these five *retrieval services* are then fed into a trend analysis *processing service*, which evaluates and normalizes magnitudes of each data category. These impact statistics are sorted by order of greatest impact and combined into a composite score. The composite score, distinct trend direction and magnitudes, and raw retrieval service outputs are then passed back to the LLM core as context. Additionally, a chart signals processing service crossed analyzes market value and fire data to highlight trends and interactions between these variables over time.

3.2.2 Chat Endpoint

The chat endpoint passes user query JSON report data back to the LLM core for analysis. Upon receiving the natural language interpretation from the backend, this endpoint then sends natural language report interpretation and definition assistance back to the front-facing interface.

3.3 Home Risk Client Class: LLM-core

The Home Risk Client class is a client of the LLM-proxy class and defines the fundamental logic to draw natural language risk insights for home buyers and owners. The risk client is initialized to use the GPT 4o-mini model under a single “LA-homeowner” session ID, with its k-value – that is, conversation history retention – set at k=5. While relatively low, this k-value was found to sufficiently meet system needs, which do not often require conversation history. The primary report generates static insights (no need for multi-turn interaction), and the interactive chat is designed to explain specific insights in the report rather than support long or continuous conversation with the user.

The backend handles the holistic, qualitative analysis of all sources, and transforms these various inputs into insights that succinctly prioritize, relate, and summarize risks and impacts.

3.3.1 RAG Information Retrieval

The risk client analysis methods support retrieval methods for static resources used to guide qualitative risk analysis and recommendations for a given property. The RAG database is initialized to contain FAIR Plan insurance policy [4] and relevant wildfire impact documentation [5], and CAL FIRE home hardening and retrofit guidelines [6].

The RAG threshold – that is, the minimum relevance needed to qualify for retrieval – was set to 0.3. This was kept relatively low since the set of sources were generally useful collectively and supported one another. The six closest matching vectors from the RAG database were used to inform responses (i.e.: rag k=6); this value was also adjusted a bit throughout development process to find a value that allowed answers to draw from enough information. Lastly the temperature was set to 0.2, to preserve technical, objective insights.

Notably, RAG context was not used at all to generate the Report Assistant chatbot insights; this was used only to generate the report itself. The chatbot used only the JSON formatted report data, supporting its purpose to explain and define risk insights rather than generate new analysis or speculative insights.

3.3.2 Structured Output through Tailored Prompting

The LLM output is guided by two distinct roles: a wildfire climate risk advisor and a data explainer embedded in a risk report tool.

The risk advisor draws from all available data sources and draws comprehensive insights across mitigation action data and environmental, home value, insurability, and structural risks. The risk advisor role is further split into two distinct roles: the homeowner advisor and the prospective homebuyer advisor. These advisors receive and draw insights from largely equivalent information sources, but the framing and categorization of their analysis varies according to user role. Broadly, the *homeowner* advisor focuses on generating actionable insights and objective information about the user's existing risks. The *homebuyer* advisor frames its analysis around key insights that allow prospective buyers to draw comparisons and weigh downstream financial risks of wildfires across different properties and zones.

The purpose of the report assistant is *not* to apply any additional insights beyond those generated by the respective risk advisors. Rather, the report assistant system prompt frames all responses from the perspective of a data explainer strictly within the generated report context. The assistant may aid users interactively by providing definitions, descriptions, methodology explanations, and factual information that is directly relevant to the users' interpretation of the report. The report assistant has strict guardrails that prevent speculation about future risks, subjective judgements, re-computation of quantitative data, speculative comparison, or unrelated topics.

4 Evaluation Methods and Results

4.1 System Level Evaluation

The system was evaluated through a structured pipeline applied across two development iterations, testing the full /analyzeFireRisk endpoint on ten predefined cases: five homeowner and five homebuyer scenarios spanning a range of geographic risk profiles. Each generated report was assessed through two mechanisms. First, automated rule-based checks verified structural compliance, including the presence of required fields, confidence validity, non-empty report sections, sufficient risk signal count, and the absence of cross-role content leakage. Second, an LLM-as-judge rubric scored each report from 1–5 across areas of role fit, data grounding, specificity, confidence calibration, and safety and hallucination avoidance. The second iteration extended the rubric with three additional RAG-specific dimensions of retrieval relevance, grounding faithfulness, and source relevance.

Across both iterations, automated checks passed at a rate of 95/105. All ten failures were concentrated in buyer-path cases, which included homeowner-style mitigation recommendations and structural property language in confidence explanations. Data grounding was the strongest-performing dimension in both iterations, averaging 4.9/5 in iteration 1 and 4.7/5 in iteration 2, reflecting reliable integration of structured data services across hazard zone classification, fire history, Zillow market indices, FAIR Plan coverage, and DINS damage records. Homeowner cases performed well overall in iteration 1, with Topanga, Malibu, and Paradise passing all rubric dimensions. The second iteration introduced three RAG-specific rubric dimensions, all of which scored 1/5 across all cases, reducing overall pass rates to zero. This decline is attributed primarily to a limitation of the LLM-as-judge approach because of the RAG, which consisted of readable documents such as home insurance policy PDFs and CAL FIRE home hardening guidelines. The judge was unable to reliably distinguish whether report language was grounded in retrieved chunks or generated from prior training.

4.2 User-Level Evaluation

4.2.1 User Studies and Feedback Surveys

User feedback was collected through a post survey and across multiple informal user studies conducted throughout the development process. The survey gathered background information, overall experience and trust ratings on a 1–5 scale, and a self-reported assessment of information sufficiency. Participants also rated individual system components, including the header signals overview, mitigation steps, structural damage information and comparison, home value trends, insurance coverage and risk analysis, and the chatbot feature, across categories ranging from very useful to potentially misleading or difficult to understand. Open-ended questions asked about the most and least useful aspects of the system, interface usability observations, and general feedback.

Qualitative feedback showed that participants generally found the system intuitive and valued that the report not only showed mitigation steps, but also surfaced financially decision-relevant information, particularly around insurance availability and home value trends. The FAIR Plan explanation was highlighted as especially useful given low baseline familiarity with the program among general users. The risk visualization and report summary were positively received, and the chatbot feature was noted as a valuable complement to the static report.

4.2.2 Randomized Controlled Experiment

A randomized controlled experiment was conducted with 6 participants to assess whether the system meaningfully improves users' understanding of wildfire risk, with a focus on the homeowner's path. Participants were assigned to the role of an LA homeowner living in North Topanga Canyon Boulevard, Topanga, Los Angeles County, an area typically considered high risk due to dense vegetation and limited evacuation due to proximity with the Santa Monica Mountains. Users were also given a set of property characteristics: asphalt roof construction, no eaves, and stucco siding. All users received an introduction of the same goal: self-identify wildfire risks and possible courses of action, accounting for environmental, financial, and property-specific factors. A brief general wildfire information introduction was provided to all participants prior to the task.

Users were then placed in one of two groups: those which used our Wildfire Risk Assessment Tool (Group A) and those who used their preferred LLM Chatbot/open web search (Group B). All participants were given up to 15 minutes to investigate financial, home-specific, and environmental risk information for the fixed property address with the standardized home conditions. Although they were able to ask questions during the experiment, participants in group A received no explicit instructions on how to use the wildfire tool; this was intended to consider the tool's intuitive functionality.

4.2.2.1 Data Collection

Pre-study surveys collected demographic information relevant to potential confounds, including prior California residency, familiarity with FAIR Plan and state guidelines, age, technical experience, role experience (prior home buying or ownership), and disciplinary background. No significant disparities with respect to previous experience or background knowledge were found, so users were assigned to their experiment groups randomly. The majority of participants reported high usage/familiarity with LLMs and AI tools, with 66% of users rating their usage at a 5 on a 5-point scale. These participants were also primarily young (83% under 30) and reported favorable opinions of generative AI in general.

Post-task assessment evaluated participants across two dimensions. Knowledge assessment measured understanding of financial, home-specific, and environmental risk through a combination of objective questions and open-ended responses, which were classified for level of detail, correctness, and cross-category cohesion. Action assessment asked

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participants to identify one to three concrete next steps given their assigned role, evaluating the degree to which information gained translated into actionable, role-appropriate decisions. Participants in AI-assisted groups additionally completed a user experience survey rating specific system feature by perceived usefulness.

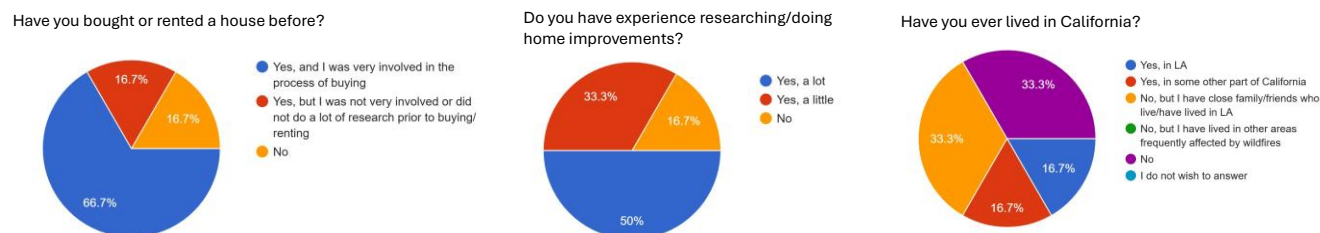


Figure 4: Pre-survey results for relevant background

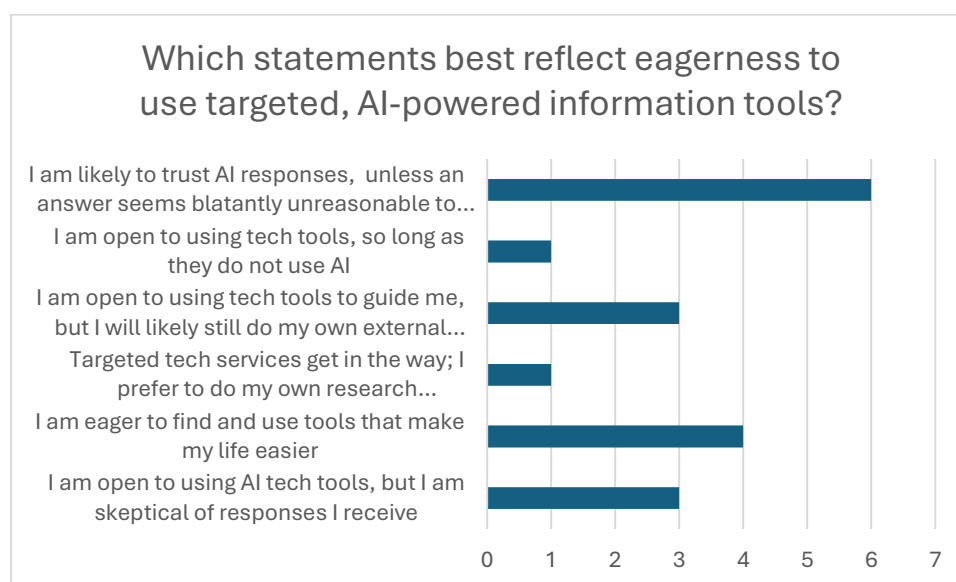


Figure 5: The above figure shows number of participants that agreed with statements about trust and general view of AI tools. Most users agreed with statements reflecting high trust in AI and favorable views of targeted AI services

4.2.2.2 Metrics

Users in both groups completed information assessments as well as self-reported experience structured and semi-structured feedback. For multiple-choice, 5-point ratings, and multi-select, results were assessed according to average ratings or number of users that selected options, respectively. Open-ended responses were collected for four distinct questions that asked users to identify and describe 1) major financial risks, 2) physical property characteristics and related vulnerabilities, 3) risk factors that could be changes, 4) tangible and actionable next steps.

Open-ended assessment responses were then encoded on 20-point scale, with up to five points awarded for each of the following four evaluation categories: accuracy, specificity,

reasoning, and comprehensiveness. These encodings were first computed manually. To reduce scoring biases, the responses, context, and criteria were fed into Claude Sonnet 4.6, and the LLM-assigned scores were averaged with the manually assigned scores to generate a composite score in across each category. The mean absolute difference between human and LLM-generated scores was computed and compared as seen in Figure 7 below. The scoring was found to be relatively consistent, with disagreement frequently averaging below one point.

Rater Scoring Alignment:

Mean Absolute Difference (MAD) Across Human and LLM Encoding

	Accuracy MAD	Specificity MAD	Reasoning MAD	Comprehensiveness MAD	Overall MAD
Q1: financial risk	1.33	1.17	0.67	1.17	1.08
Q2: physical chars	0.5	0.5	0.17	0.83	0.5
Q3: Action/risks	0.5	0.83	0.67	1.33	0.83
Q4: Concrete Steps	1	1.33	1.5	2	1.45
	0.8325	0.9575	0.7525	1.3325	0.965

Figure 7: The above figure shows the mean absolute differences between manual, human encodings and LLM-generated encodings. Each user short-answer response was scored on a 5-point scale for each of the four established categories: accuracy, specificity, reasoning, and comprehensibility.

4.2.2.3 RCT Results

Self-Reported System-Trust Ratings

Overall, users interacting with the targeted Wildfire Risk Assessment system reported slightly higher overall trust of information received, averaging at 4.67 on a 5-point scale, while the groups that used general AI tools reported an average trust of 3.67. Nevertheless, neither group expressed significant distrust of the information they received.

Feature Usefulness

Two out of three Group A users reported finding the interactive report assistant chatbot somewhat useful, while the remaining user tried it but did not continue to use it. Only one Group A user reported exporting the report to an independent tab. This user expressed positive feedback for this feature; however, the other Group A users did not report using this feature at all.

Control Variables

Users in Group B were given the freedom to choose the general purpose LLM/internet tools of their choice. This was done to control for potential discomfort with less familiar systems or research methods. Users in Group B reported using ChatGPT 5.3 (General Purpose Version), Google web search, and Gemini 3 (Fast Mode). Notably, both LLM models are more potent models than the 4o-mini, which was used to power the Wildfire Risk Assessment Tool. There

was not a substantial or consistent difference in the time used, though this was observed for both groups.

Additionally, users were observed to have varying levels of comfort prompting and using general purpose models/ web search. Some users expressed frustration with the general-purpose systems and were less willing to continue prompting LLMs. This resulted in consistent relative score rankings between users in Group B, as seen in figure 8.

Avg Scores Per User	Percent Total Possible Points					
	GROUP A			GROUP B		
	User1	User2	User3	User4	User5	User6
Q1: financial risk	65%	50%	78%	33%	73%	93%
Q2: physical property chars	50%	58%	45%	25%	53%	100%
Q3: Action/risks	83%	55%	68%	20%	38%	98%
Q4: Concrete Steps	83%	70%	75%	30%	45%	90%
Total	70%	58%	66%	27%	52%	95%

Figure 8: Shows percentage of total possible points scored in each category of assessment. The maximum score of 80 reflects 5-point ratings across all evaluation categories, for all question categories. The percentage was computed using the average score between the human and LLM response encoding

Post-Experiment Information Assessment and Self-Reported Feedback:

Results and Discussion

The most significant observed results across Groups A – which used the Wildfire System Tool – and Group B, which did not, was the consistency in responses within the group. While the average scores do not show a significant distinction between the treatment and control groups, inspection of the raw data reveals that this is largely due to group B scores being inflated by a single user (user F), who consistently scored very highly across all categories. Notably, this user in particular expressed high familiarity and comfortability using the LLM system.

As seen in Figure 8, users in Group A had more variance in their scoring, with percentages computed relative to the maximum score of 80 raw points. The total scores for Group A ranged from 58%-70% while the total scores for Group B ranged 27% - 95%. This significantly higher range suggests that the targeted Wildfire Assessment tool may be less dependent on individual LLM experience, though a larger experiment sample size would need to be run to confirm this observed behavior quantitatively. Additionally, multiple users in the control Group B – including the highest-performing user F – expressed that the information they received from general-purpose LLMs felt too broad and required extensive prompting. Some users expressed visible frustration with the general-purpose LLMs, reporting that the LLMs provided too much information, even when prompted for specific charts or bullet points.

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Avg Deviation from Mean Group Score

	Stdev Accuracy		Avg Specificity		Avg Reasoning		Avg Comprehensibility	
	Group A	Group B	Group A	Group B	Group A	Group B	Group A	Group B
Q1: financial risk	1.6	0.98	0.52	1.6	1.17	1.97	1.32	1.63
Q2: physical property chars	0	1.83	0.63	1.83	0.55	1.86	0.82	1.83
Q3: Action/risks	0.83	1.9	1.17	2.07	0.82	1.75	1.41	1.97
Q4: Concrete Steps	0.82	1.67	1.55	1.86	1.22	1.21	1.86	1.63
	0.8125	1.595	0.9675	1.84	0.94	1.6975	1.3525	1.765

Average Scores Per Eval Category

	Avg Accuracy		Avg Specificity		Avg Reasoning		Avg Comprehensibility	
	Group A	Group B	Group A	Group B	Group A	Group B	Group A	Group B
Q1: financial risk	3.83	4.17	3.33	2.17	2.83	3.5	2.83	3.33
Q2: physical property chars	3	2.83	3	3.16	1.5	2.67	2.67	3.17
Q3: Action/risks	4.5	3	3.83	2.33	3	2.33	1.33	2.67
Q4: Concrete Steps	4.33	3	4	2.67	3.5	2.67	3.33	2.67
	3.915	3.25	3.54	2.5825	2.7075	2.7925	2.54	2.96

Figure 6: (Top) The average deviation from mean group score reflects how similar scores were for users within the same group (treatment or control). The treatment Group A consistently demonstrates a lower deviation across all evaluation categories, while deviation for Group B is almost doubled across every category. (Bottom) The average scores per evaluation are neither consistently nor significantly different, although accuracy and specificity favor Group A slightly.

Users who used the AI-powered Wildfire Risk Assessment system rated the system favorably overall, with an average 4.5 rating on a 5-point scale. Nevertheless, neither users in Group A nor in Control Group B rated above a three (out of 5) when asked how confidently they could explain the risks to a peer. Moreover, some Group A system users expressed continued limitations in understanding *why* certain factors caused risk, despite feeling like they had actionable next steps. Other Group A system-users expressed desire for more property-specific insights beyond zip-code level risk. This included information relating to natural barriers, proximity to emergency units, and information relating to evacuation planning/routes.

RCT Conclusions

While the scope of the randomized controlled experiment does not allow definite conclusions with respect to the system effectiveness, the observed results suggest promising insights with respect to both the functionality and accessibility of the specialized Wildfire Risk Assessment system. The favorable difference in consistency suggests there may be a significant advantage in the average/baseline quality and scope of information retrieved by the system across users with different levels of experience and willingness to extensively search the web or continuously prompt a general-purpose LLM. Nevertheless, the consistently high-performing LLM-user suggests that there may be remaining limitations to the *depth* of the insights generated by the specialized Risk Assessment system. This and additional limitations are described in further detail in section 5.

5 Limitations

Several system limitations exist with respect to data scope, display, and evaluation methodology, as discussed in the sections that follow.

5.1 Data Limitations

There are inherent limitations in the coverage and exhaustiveness of the data considered in the system's risk analysis. The system relies heavily on third-party data sources, including Zillow market indices, CAL FIRE hazard zone classifications, and California Department of Insurance records. This introduces dependencies on data that potentially contain coverage gaps, particularly for less-populated ZIP codes. In addition, the availability of substantial, free data is often limited to zip-code level information. Thus, report generation may fail to capture more fine-grain, property-specific insights. Moreover, while the insights of the generated reports draw from many trusted data sources across a breadth of relevant disciplines, there may still be explanatory variables that are not entirely accounted for in the risk analysis. For example, the system does not consider proximity to available evacuation routes or wildfire emergency dispatch units.

The architecture of the system—in which data is drawn from purpose-specific, often dynamic services and static RAG documentation—supports simple, systematic integration of additional data sources. Updating obsolete or outdated data is similarly simple for data queried from services; nevertheless, because the RAG database does not support clearing or removal functionality, updates to static sources may lead to database clutter, redundancy, or even conflicting data.

5.2 User Experience: Limited Interactive Capabilities

The current report is entirely static, delivering all available information at once in one modal. While the chatbot helps to make the report more interactive, user feedback consistently indicated that the volume and density of information across sections, particularly home value trends and structural damage comparisons, created cognitive overload, making it difficult for users to identify and act on the most decision-relevant insights for their role.

5.3 System Evaluation Limitations: Depth and Scope

The system evaluation is currently comprised of only ten cases, which constrains the generalizability of observed performance patterns. Similarly, the controlled experiment is subject to significant statistical limitations with only 6 participants, which largely represented similar age groups and experience with generative AI. Thus, the participant pool likely does not exhaustively reflect the broader population of homeowners and homebuyers the system is designed to serve, introducing potential bias in both knowledge assessment scores and usability feedback.

6 Future Work

Several directions for future development are identified across system design, evaluation, and user experience. On the evaluation side, the diversity of the evaluations should be expanded to a larger and more geographically diverse set of cases to improve the generalizability of performance assessments.

A key user-facing improvement is the introduction of a property comparison feature, allowing users to generate and view risk reports for multiple addresses side by side. User feedback consistently identified this as a beneficial feature, particularly for prospective buyers evaluating multiple properties simultaneously. This would complement the existing exportable report feature and enable more informed, comparative decision-making across locations, insurance risk profiles, and market trends.

Another key extension is in the continued research and improvement in the depth and breadth of the data used to inform risk insights. Existing data limitations discussed above can be addressed through further expert analysis of factors that are relevant to wildfire risk. Additionally, access to more and deeper data insights may allow the scope of the broader system to extend beyond LA Wildfires. This may involve evolving the system framework to support larger geographical areas (statewide, nationwide, global) or a more diverse range of natural disasters and climate events, including but not limited to flooding or earthquake risks.

The current system supports two user paths, homeowner and homebuyer, but future work should explore additional role-specific workflows that reflect the broader range of stakeholders affected by wildfire risk. Potential paths include journalists investigating the scope of the California wildfire insurance crisis, economists studying the relationship between fire risk and market dynamics, and policymakers designing land use or insurance interventions at the zone or regional level. Each additional path would require tailored data prioritization, prompt design, and output structure to ensure role-appropriate insights.

On the interface side, incorporating a layered report of presentation would be a promising direction for reducing information overload. It would focus on delivering high-level summaries first and then exposing more detailed information next. This would be consistent with user feedback requesting more guided and digestible experience.

7 Acknowledgements

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Appendix

Emily

Through this course, I developed a deeper understanding of how generative AI systems are built, evaluated, and deployed responsibly for social impact. Topics like prompt engineering, retrieval-augmented generation, guardrails, and LLM-as-judge evaluation were all crucial in enabling me to fundamentally understand how to design and contribute to the Wildfire Risk Analysis Tool. I was able to utilize these concepts in our final project by building a system that combined external hazard, structural, and financial data with grounded LLM outputs, reflecting the course's emphasis on agentic workflows and real-world applications. Overall, this course taught me to prioritize user experience and human-centered design, especially when building tools aimed at addressing real-world problems, ensuring that systems are trustworthy, accessible, and meaningful to the people who rely on them.

Yvonne

This course/project allowed me to broaden my understanding of the potentially positive role that generative AI can play across a variety of disciplines. Implementing this wildfire project was particularly interesting for me, as it allowed me to bridge my interests in environmental sustainability with my technical computer science/systems interests. It was awesome to be able to consult our external project mentor, Scott McCullers, who played a huge role guiding our system design and sharing his expertise/experiences working with real-world applications in the Microsoft AI for Good Lab. Hearing from various user types was additionally a very meaningful part of this project, as our developments were guided to meet

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user-needs rather than by strict technical guidelines or objectives. This was my first time performing a formal user-study experiment, and it was cool to be able to see the metrics converge towards trends and conclusions (even within the small test group size).

Coming into the course, I considered AI to be little more than a means of increasing efficiency. I was curious about the long-term sustainability of AI, as I questioned whether immediate efficiency benefits could outweigh potential long-term economic and social costs. To me, the acclaimed gains of AI often seemed to detract from or come at the expense of human skills and creativity, and—in a rapidly changing industry where adaptability, flexibility, and efficiency are often valued above all—I rarely had the time to challenge these internal perceptions in a structured or academic way. Nevertheless, this course allowed me to explore these possible costs and think more deeply about meaningful applications of generative AI. My perspective on AI broadened greatly as I learned that effective and careful use of AI can often complement rather than replace human skills and creativity, and I was able to learn about and build generative-AI-based systems designed to serve very human-centered objectives. Huge thanks to Professor Dogar for making all of that possible!